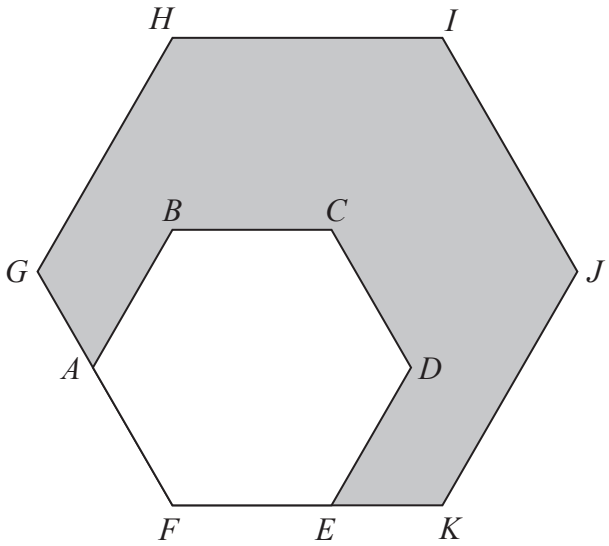
19



ABCDEF is a regular hexagon with sides of length x.
This hexagon is enlarged, centre F, by scale factor p to give hexagon *FGHIJK*.

Show that the area of the shaded region in the diagram is given by $\frac{3\sqrt{3}}{2}(p^2 - 1)x^2$

A regular hexagon comprises 6 equilateral triangles.

One triangle of side x has area $\frac{1}{2}x^2\sin 60° = \sqrt{3}x^2/4$.

Inner hexagon: $6 \times \sqrt{3}x^2/4 = 3\sqrt{3}x^2/2$.

Enlargement p multiplies area by p^2.

Shaded area = $3\sqrt{3}p^2x^2/2 - 3\sqrt{3}x^2/2$

= $(3\sqrt{3}/2)(p^2 - 1)x^2$, as required.

4 marks total
Triangle area
Inner hexagon
Outer p^2 area
Subtract to result

(Total for Question 19 is 4 marks)

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